Real-Time Hand Sign Recognition using Deep Learning

Abstract

This project presents a real-time hand sign recognition system built using deep learning and computer vision techniques. The system detects a hand region using MediaPipe and classifies gestures using a transfer learning model based on MobileNetV2. The model was trained on a custom self-collected dataset consisting of over 25,000 images across 36 classes representing numbers (0–9) and alphabets (A–Z). The system achieves approximately 99% validation accuracy and supports live webcam inference with confidence scores.

Introduction

Hand gesture recognition plays an important role in human-computer interaction and assistive communication systems. This project focuses on building an efficient and accurate real-time hand sign recognition system using deep learning. Unlike traditional image classification tasks, real-time gesture recognition introduces challenges such as lighting variation, background noise, and unstable predictions during live inference. To address these challenges, the system integrates MediaPipe for hand detection and MobileNetV2-based transfer learning for robust classification.

Tools Used

- Python – Programming language
- TensorFlow and Keras – Deep learning framework
- MobileNetV2 – Pretrained CNN model for transfer learning
- OpenCV – Real-time webcam processing
- MediaPipe – Hand detection and landmark tracking
- NumPy – Numerical operations
- Matplotlib and Seaborn – Visualization of training metrics
- Scikit-learn – Confusion matrix and classification report

Steps Involved in Building the Project

1. Dataset Creation: Collected over 25,000 hand gesture images using webcam capture under varying lighting conditions and angles. Organized images into 36 class folders.
2. Data Preprocessing: Rescaled pixel values, applied data augmentation (rotation, zoom, flipping), and automatically split the dataset into training and validation sets.
3. Model Development: Implemented transfer learning using MobileNetV2 with frozen base layers and a custom classification head.
4. Model Training: Trained the model using categorical cross-entropy loss and Adam optimizer, achieving high validation accuracy.
5. Evaluation: Generated accuracy and loss curves, confusion matrix, and classification report to analyze class-wise performance.
6. Real-Time Inference: Integrated MediaPipe to detect and crop the hand region, then performed live webcam classification with confidence percentage and prediction smoothing.

Conclusion

The project successfully demonstrates the development of a high-accuracy real-time hand sign recognition system using deep learning. By combining transfer learning with efficient hand detection, the system achieves stable and reliable predictions. The implementation highlights the importance of data preprocessing, proper evaluation metrics, and real-time optimization techniques. This project provides a strong foundation for future enhancements such as full sign language recognition, mobile deployment using TensorFlow Lite, and integration with assistive communication systems.